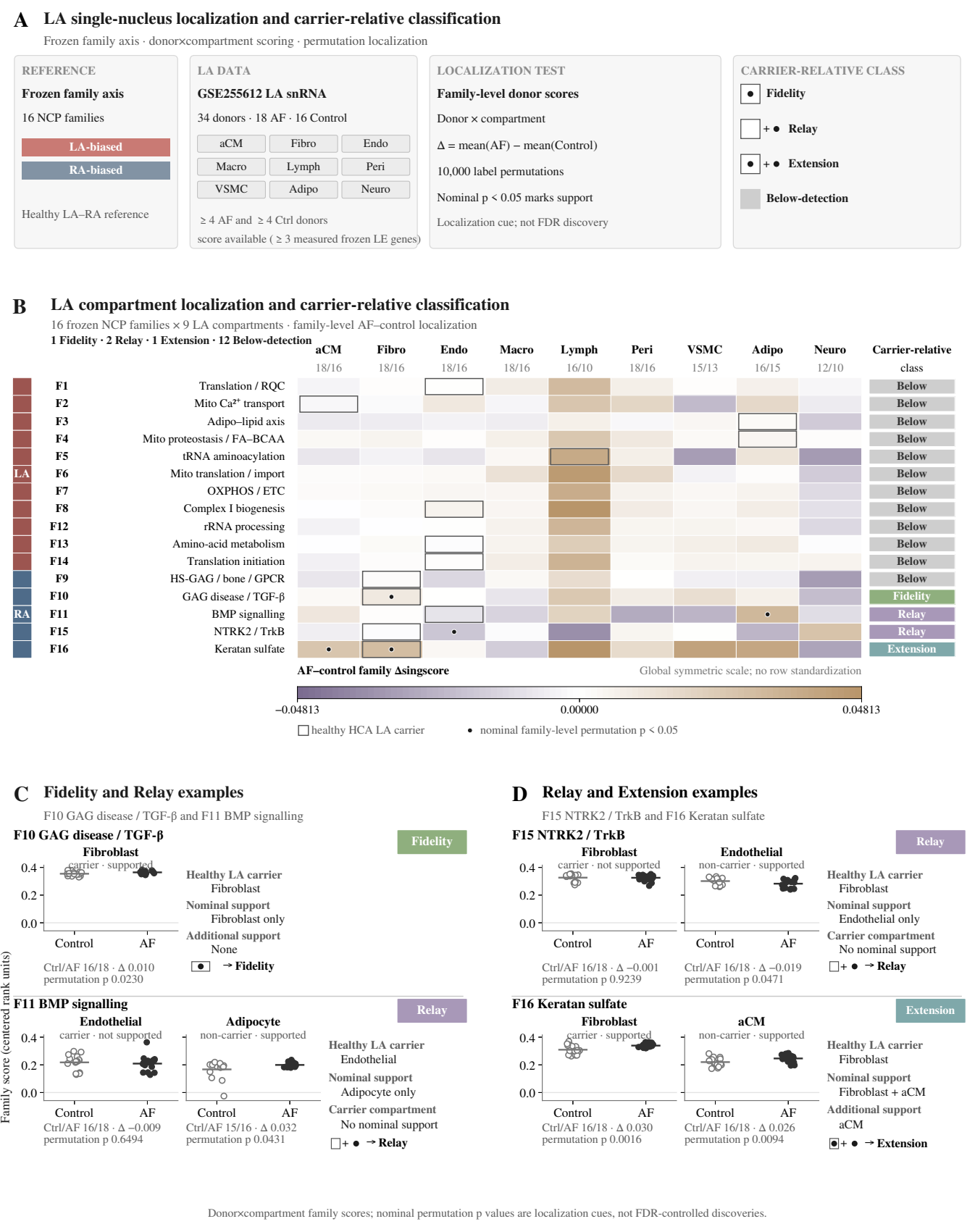

A
LA single-nucleus localization and carrier-relative classification
Frozen family axis · donor×compartment scoring · permutation localization
REFERENCE
Frozen family axis
16 NCP families
LA-biased
RA-biased
Healthy LA–RA reference
LA DATA
GSE255612 LA snRNA
34 donors · 18 AF · 16 Control
aCM
Fibro
Endo
Macro
Lymph
Peri
VSMC
Adipo
Neuro
≥ 4 AF and ≥ 4 Ctrl donors
score available (≥ 3 measured frozen LE genes)
LOCALIZATION TEST
Family-level donor scores
Donor × compartment
Δ = mean(AF) − mean(Control)
10,000 label permutations
Nominal p < 0.05 marks support
Localization cue; not FDR discovery
CARRIER-RELATIVE CLASS
Fidelity
+ Relay
+ Extension
Below-detection
B
LA compartment localization and carrier-relative classification
16 frozen NCP families × 9 LA compartments · family-level AF–control localization
1 Fidelity · 2 Relay · 1 Extension · 12 Below-detection
aCM 18/16
Fibro 18/16
Endo 18/16
Macro 18/16
Lymph 16/10
Peri 18/16
VSMC 15/13
Adipo 16/15
Neuro 12/10
Carrier-relative class
LA
RA
F1 Translation / RQC Below
F2 Mito Ca2+ transport Below
F3 Adipo–lipid axis Below
F4 Mito proteostasis / FA–BCAA Below
F5 tRNA aminoacylation Below
F6 Mito translation / import Below
F7 OXPHOS / ETC Below
F8 Complex I biogenesis Below
F12 rRNA processing Below
F13 Amino-acid metabolism Below
F14 Translation initiation Below
F9 HS-GAG / bone / GPCR Below
F10 GAG disease / TGF-β Fidelity
F11 BMP signalling Relay
F15 NTRK2 / TrkB Relay
F16 Keratan sulfate Extension
AF–control family Δsingscore
Global symmetric scale; no row standardization
−0.04813
0.00000
0.04813
healthy HCA LA carrier
nominal family-level permutation p < 0.05
C
Fidelity and Relay examples
F10 GAG disease / TGF-β and F11 BMP signalling
F10 GAG disease / TGF-β
Fidelity
Fibroblast
carrier · supported
Control
AF
Healthy LA carrier
Fibroblast
Nominal support
Fibroblast only
Additional support
None
→ Fidelity
Ctrl/AF 16/18 · Δ 0.010
permutation p 0.0230
F11 BMP signalling
Relay
Endothelial
carrier · not supported
Adipocyte
non-carrier · supported
Healthy LA carrier
Endothelial
Nominal support
Adipocyte only
Carrier compartment
No nominal support
→ Relay
Ctrl/AF 16/18 · Δ −0.009
permutation p 0.6494
Ctrl/AF 15/16 · Δ 0.032
permutation p 0.0431
Family score (centered rank units)
D
Relay and Extension examples
F15 NTRK2 / TrkB and F16 Keratan sulfate
F15 NTRK2 / TrkB
Relay
Fibroblast
carrier · not supported
Endothelial
non-carrier · supported
Healthy LA carrier
Fibroblast
Nominal support
Endothelial only
Carrier compartment
No nominal support
→ Relay
Ctrl/AF 16/18 · Δ −0.001
permutation p 0.9239
Ctrl/AF 16/18 · Δ −0.019
permutation p 0.0471
F16 Keratan sulfate
Extension
Fibroblast
carrier · supported
aCM
non-carrier · supported
Healthy LA carrier
Fibroblast
Nominal support
Fibroblast + aCM
Additional support
aCM
→ Extension
Ctrl/AF 16/18 · Δ 0.030
permutation p 0.0016
Ctrl/AF 16/18 · Δ 0.026
permutation p 0.0094
Donor×compartment family scores; nominal permutation p values are localization cues, not FDR-controlled discoveries.